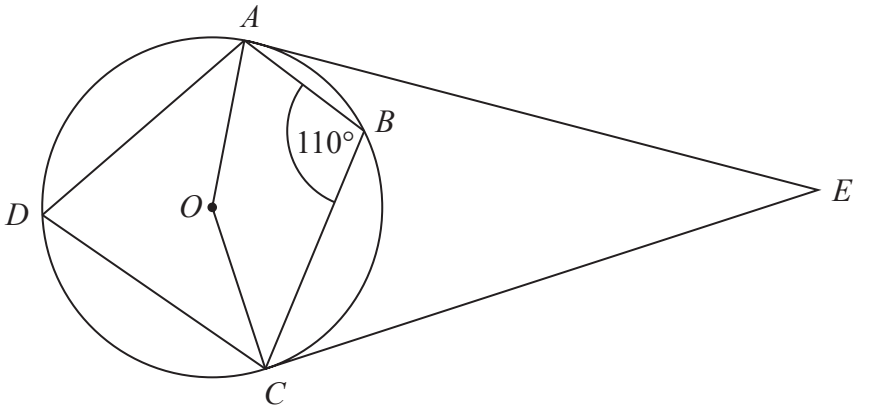
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15



A, B, C and D are points on the circumference of a circle, centre O.
AE and CE are tangents to the circle.

Angle $ABC = 110°$

Work out the size of angle AEC.
You must show all your working.

OA and OC meet their tangents at right angles.

1 mark
Radius meets tangent at 90°.

So angle OAE = angle OCE = 90°.

1 mark
Use the same arc.

Reflex angle AOC = 2 × 110° = 220°

(centre angle is twice the circumference angle).

Minor angle AOC = 360° – 220° = 140°.

1 mark
Quadrilateral angle sum.

In quadrilateral AOCE:

angle AEC = 360° – 90° – 90° – 140° = 40°.

1 mark
40°.

.....°

(Total for Question 15 is 4 marks)